

AI Internship at Peropytx

An Internship report for the evaluation and partial fulfillment of the requirement for
the award of the degree

B. TECH. CSE (SPECIALIZATION IN MACHINE LEARNING)



Submitted By -

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(215/UCM/017)

UNDER THE SUPERVISION OF

Ms. Manjari Gangwar

UNIVERSITY SCHOOL OF INFORMATION AND COMMUNICATION

TECHNOLOGY GAUTAM BUDDHA UNIVERSITY

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UTTAR PRADESH INDIA

March 2025



SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY
GAUTAM BUDDHA UNIVERSITY, GREATER NOIDA, 201312, U. P., (INDIA)

Candidate's Declaration

I, hereby, certify that the work embodied in this industrial training report entitled “**Internship at Peroptyx**” in partial fulfillment of the requirements for the award of the Degree of **B. Tech. CSE with specialization in Machine Learning** submitted to the School of Information and Communication Technology, Gautam Buddha University, Greater Noida is an authentic record of my own work carried out under the supervision of Ms. Manjari Gangwar, department of Computer Science and Engineering, University School of Information and Communication Technology. The matter presented in this report has not been submitted in any other University / Institute for the award of any other degree or diploma. Responsibility for any plagiarism related issue stands solely with me.

NAME	ROLL NO	SIGNATURE
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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief. However, responsibility for any plagiarism related issue solely stands with the student.

Signature of the Supervisor:
Name: Ms. Manjari Gangwar

Date: 5th March, 2025
Place: Gautam Budh Nagar

Acknowledgement

I would like to express my sincere gratitude to all those who helped me throughout the course of this internship. It would not have been possible without the support and assistance of many individuals and organizations, and I am deeply thankful for their contributions.

First and foremost, I am highly indebted to Ms. Manjari Gangwar, my supervisor, department of Computer Science and Engineering, University School of Information and Communication Technology. Her invaluable guidance, constant supervision, and encouragement throughout the internship were essential to my learning experience. Her support in providing crucial information regarding the training was instrumental in my success.

I would also like to thank Dr. Arun Solanki, the Head of the Department of Computer Science and Engineering, for his support, which greatly helped me in completing this training.

Lastly, I would like to express my gratitude towards my friends and family members for their constant cooperation, understanding, and encouragement, which played a key role in the successful completion of this internship.

Abstract

This report details my internship experience as an AI Intern at Peroptyx, which began on March 3rd, 2025. Peroptyx specializes in providing high-quality machine learning (ML) training data and model evaluation solutions, particularly for location-based services and AI-driven applications. The internship offered hands-on experience in AI model evaluation, data annotation, and improving the accuracy of ML models used in real-world applications.

During the internship, I worked on several key projects, including evaluating AI-generated search results, annotating data for natural language processing (NLP) models, validating geospatial data for online maps, moderating AI-generated content, and analyzing ML model performance. These projects focused on enhancing AI models' relevance, fairness, and accuracy by ensuring high-quality input data and robust evaluation methodologies.

Throughout my role, I used various AI tools and platforms to analyze data, assess ML model performance, and contribute to optimizing AI-driven products. This experience helped me strengthen my technical skills in data annotation, ML evaluation, and AI bias detection while also improving my analytical and problem-solving abilities.

The internship provided valuable exposure to industry best practices in AI model evaluation and data quality management. Additionally, it allowed me to develop essential professional skills such as time management, adaptability, and attention to detail in remote work settings.

This report summarizes my key learnings, contributions, and the impact of my work at Peroptyx. It also highlights the significance of high-quality training data in improving ML systems and provides insights into the role of AI model evaluation in optimizing AI-driven applications.

Offer Letter



Date: 25th February 2025

Subject: **Appointment Letter for 6-Month Internship**

Dear Parnika Singh,

We are pleased to offer you the position of AI Intern at Peroptyx as part of your academic internship program. This internship will provide you with valuable hands-on experience in AI model evaluation, data annotation, and machine learning applications.

Internship Details:

- Position: AI Intern
- Start Date: March 3, 2025
- Duration: 6 Months
- Location: Remote/Noida Office
- Working Hours: Flexible, up to 20 hours per week
- Stipend: Rs. 15,000

Roles and Responsibilities:

As an AI Intern at Peroptyx, your key responsibilities will include:

- Evaluating AI-generated search results and providing quality feedback.
- Conducting geospatial data validation for improving map-based AI models.
- Assisting in content moderation and bias detection in AI-driven platforms.
- Contributing to data annotation and model training for ML-based solutions.
- Collaborating with team members to enhance AI evaluation methodologies.

Terms and Conditions:

- You are expected to maintain confidentiality regarding all company data and processes.
- You must adhere to company guidelines and complete assigned tasks on time.
- This internship does not guarantee full-time employment but serves as an opportunity for skill development and professional growth.

Please confirm your acceptance by signing and returning a copy of this letter by 28th February. If you have any questions, feel free to contact us at recruitment@peroptyx.com / manik.hr@peroptyx.com.

We look forward to welcoming you to the Peroptyx team and are excited about the contributions you will make during your internship.

Best Regards,

Mr. Manik Seth
HR Department
Peroptyx
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Phone: +917982576883

Contents

Title	I
Candidate’s Declaration	II
Acknowledgement	III
Abstract	IV
Offer Letter	V
Introduction.....	7
1. About Peroptyx	7
2. Objective of the Internship.....	8
Internship Role and Responsibilities.....	9
1. Job Description	9
2. Key Responsibilities	10
Projects.....	13
1. Evaluation of AI Generated Search Results.....	13
2. Geospatial Data Validation for Maps	14
3. Content Moderation and Bias Detection.....	14
Skills Gained and Contributions	16
1. Technical Skills.....	16
2. Professional and Soft Skills	17
Conclusion and Recommendations.....	18
1. Summary of Experiance.....	18
2. Futue Scope and Career Impact	19
3. Recommendation for Future Interns	19

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have become integral to modern digital services, driving advancements in search engines, recommendation systems, mapping applications, and content moderation. However, the effectiveness of these AI systems heavily depends on the quality of their training data and evaluation processes. Peroptyx, a leading provider of ML training data and model evaluation solutions, plays a crucial role in ensuring that AI models are accurate, unbiased, and contextually relevant. Through this internship, I had the opportunity to work on AI model evaluation and data annotation, contributing to the refinement of machine learning systems. This report provides an overview of my experience at Peroptyx, detailing the projects undertaken, skills gained, and the overall impact of my contributions.

1. About Peroptyx

Peroptyx is an AI solutions provider specializing in location-based machine learning training data, data annotation, and model evaluation. Founded in 2018 and headquartered in Castlebar, County Mayo, the company collaborates with global technology firms to enhance AI-driven applications by providing high-quality data insights. The company focuses on improving AI models used in search engines, maps, speech recognition, and content moderation.

One of Peroptyx's key strengths lies in its commitment to delivering precise, unbiased, and well-annotated datasets, ensuring that AI models learn from high-quality data. Their AI evaluation services enable businesses to refine ML models for better user experience and regulatory compliance. Peroptyx's platform integrates authentication, onboarding, use-case education, and performance analytics to enhance the efficiency of data annotation and evaluation. My internship at Peroptyx involved hands-on experience with these processes, giving me insight into industry practices and the importance of high-fidelity training data.

2. Objectives of the Internship

The primary goal of this internship was to gain practical experience in AI model evaluation and data annotation while understanding the importance of high-quality training data in ML systems.

Through this internship, I aimed to develop technical expertise in AI evaluation, improve analytical skills, and contribute to real-world AI projects.

Key objectives included:

- a) Understanding AI model evaluation techniques and best practices.
- b) Working with real-world datasets to improve ML model accuracy and relevance.
- c) Gaining hands-on experience with annotation tools and evaluation methodologies.
- d) Enhancing problem-solving and critical thinking skills in AI applications.
- e) Collaborating with industry professionals to understand how AI models are refined.

This internship provided valuable exposure to ML training data pipelines and the role of human evaluation in optimizing AI systems. By working on various projects, I was able to apply theoretical knowledge in a practical setting while learning how AI models are continuously improved for better performance and user experience.

Internship Role and Responsibilities

During my internship at Peroptyx as an AI Intern, I was responsible for evaluating machine learning (ML) models, annotating datasets, and improving AI-driven applications. The role required a keen analytical approach to assess AI-generated outputs, ensuring that they were accurate, contextually relevant, and aligned with predefined quality standards. My work primarily focused on optimizing AI models used in search engines, mapping services, content recommendations, and natural language processing (NLP) applications.

This internship provided an excellent opportunity to apply theoretical knowledge of artificial intelligence and machine learning in a real-world industry setting. By working with large-scale datasets, I gained insight into how AI models are trained and refined to enhance user experience and performance. Additionally, I developed a better understanding of data annotation techniques, bias detection, and quality assessment methods in AI model evaluation. The following sections outline the specifics of my job description and key responsibilities.

1. Job Description

As an AI Intern at Peroptyx, my role revolved around data annotation, AI model evaluation, and quality assessment of machine learning systems. The primary focus of my work was to assess AI-generated content for accuracy, relevance, and compliance with established guidelines. Additionally, I contributed to improving machine learning models by providing high-quality training data, which helped in refining AI applications across various domains.

One of the core aspects of my role was evaluating AI-driven outputs, ensuring that search engines, virtual assistants, and recommendation systems provided users with the most relevant and useful results. This required detailed analysis, critical thinking, and adherence to quality benchmarks set by the company.

Key aspects of the job included:

- a) **AI Model Evaluation:** Assessing AI-generated content, including search engine results, advertisements, geospatial data, and text-based recommendations, to ensure alignment with user intent.
- b) **Data Annotation for Machine Learning:** Tagging and categorizing text, images, and maps to improve the training datasets used for ML models.
- c) **Bias Detection and Content Moderation:** Identifying misleading or inappropriate content, ensuring that AI-generated responses are accurate, unbiased, and compliant with platform policies.
- d) **Performance Review of AI Models:** Comparing different versions of ML models to identify improvements and highlight areas needing further refinement.
- e) **Reporting and Feedback:** Documenting findings, observations, and areas for improvement in AI models, contributing to the ongoing enhancement of machine learning applications.

This position provided a hands-on experience in the intersection of artificial intelligence, machine learning, and human-in-the-loop evaluation. Through daily tasks, I gained a comprehensive understanding of data annotation processes, quality control methods, and AI-driven content optimization strategies.

2. Key Responsibilities

Throughout my internship, I was assigned multiple tasks that required a strong analytical mindset, attention to detail, and the ability to interpret AI-generated outputs accurately. These responsibilities allowed me to develop essential skills in AI evaluation, data annotation, and ML model performance assessment.

2.1 AI Model Evaluation and Performance Analysis

One of the primary aspects of my role involved evaluating AI-generated search results, personalized recommendations, and online advertisements. I was responsible for:

- a) Reviewing search engine results and assessing their accuracy, relevance, and ranking effectiveness.

- b) Comparing different versions of AI models to measure performance improvements and highlight errors.
- c) Ensuring search queries return meaningful and high-quality results by analyzing search intent and contextual appropriateness.
- d) Providing detailed feedback to improve machine learning algorithms.

This responsibility helped me understand how AI models process and rank information and the significance of human intervention in training AI for optimal performance.

2.2. Data Annotation for Machine Learning Training

Training data is crucial for AI models to learn and improve, and my role involved preparing high-quality datasets through annotation and classification. I worked on:

- a) Tagging, categorizing, and labeling large datasets, including text, images, and geospatial data.
- b) Ensuring that the annotated data was precise, consistent, and met industry standards for ML model training.
- c) Assisting in improving natural language processing (NLP) applications by evaluating text-based AI outputs.

By participating in data annotation, I gained firsthand experience in how AI models are trained, fine-tuned, and optimized for real-world applications.

2.3 Bias Detection and Content Moderation

AI models can sometimes generate biased, misleading, or inappropriate content, making content moderation a critical part of AI evaluation. My tasks in this area included:

- a) Identifying and flagging harmful, misleading, or low-quality content in AI-generated responses.
- b) Ensuring that search results, advertisements, and recommendations were free from bias and aligned with ethical AI principles.
- c) Reviewing AI outputs for compliance with company policies and content guidelines.

This experience provided me with insight into AI ethics, fairness, and the impact of human oversight in AI-driven decision-making.

2.4 Search Relevance and Personalization

Another major component of my responsibilities was assessing search relevance and personalization algorithms. This involved:

- a) Reviewing AI-generated search results, ad placements, and personalized content recommendations for accuracy and user intent.
- b) Evaluating how AI systems tailor search results based on location, search history, and user preferences.
- c) Identifying inconsistencies in search rankings and recommending improvements.

This work deepened my understanding of how AI-driven personalization works and its impact on user engagement, experience, and satisfaction.

2.5 Documentation, Reporting, and Team Collaboration

To ensure the AI models continually improved, it was important to document findings, report inconsistencies, and provide constructive feedback. This involved:

- a) Preparing detailed reports on AI model performance and data quality issues.
- b) Participating in discussions with industry professionals to improve AI systems.
- c) Collaborating with AI teams to suggest enhancements for search algorithms and content ranking models.

Through this, I improved my ability to communicate complex technical issues effectively and gained experience in structured AI performance analysis.

Projects Undertaken

During my internship at Peroptyx, I had the opportunity to work on various projects related to AI evaluation, data annotation, and machine learning performance improvement. These projects involved assessing search relevance, geospatial data accuracy, and content moderation to enhance the overall efficiency of AI-driven applications. Each project provided valuable insights into how AI models function, their strengths, and areas needing improvement. The following sections detail the key projects I contributed to during my internship.

1. Evaluation of AI-Generated Search Results

This project focused on assessing the accuracy, relevance, and ranking of search results produced by AI-driven search engines. Since search engines rely on machine learning algorithms to retrieve and rank information, it is crucial to ensure they return meaningful and contextually appropriate results to users.

My responsibilities in this project included:

- a) Evaluating search engine results by assessing whether the retrieved web pages matched the user's query intent.
- b) Comparing AI-generated search rankings to determine if higher-ranked results were more relevant than lower-ranked ones.
- c) Identifying misleading or outdated information and providing feedback for model improvements.
- d) Testing search personalization algorithms by analyzing how AI systems adapted results based on user preferences and location.

Through this project, I gained a deep understanding of search engine optimization (SEO), natural language processing (NLP), and AI-driven personalization techniques. The work also reinforced the importance of human-in-the-loop AI evaluation to refine search engine performance. Additionally, this experience highlighted the challenges in optimizing AI-driven search algorithms while maintaining fairness and relevance for diverse user queries.

2. Geospatial Data Validation for Maps

Accurate geospatial data is essential for navigation applications, mapping services, and location-based AI models. This project involved evaluating geospatial datasets, verifying location accuracy, and ensuring map-based AI recommendations were reliable.

Key tasks I performed in this project:

- a) Reviewing AI-generated mapping data to check for discrepancies in location names, addresses, and landmarks.
- b) Validating geospatial coordinates to ensure they matched real-world locations.
- c) Assessing navigation route recommendations for correctness and efficiency.
- d) Identifying anomalies in mapping services, such as incorrect road names, missing landmarks, or inaccurate route estimations.

This project significantly improved my understanding of geospatial data science, map-based AI models, and the role of human evaluation in training location intelligence systems. Moreover, it highlighted the importance of ensuring high precision in AI-driven geospatial predictions, as even small errors in location data can lead to incorrect navigation and user dissatisfaction.

3. Content Moderation and Bias Detection

AI models are widely used for content recommendation, moderation, and filtering across various platforms. However, biased or inappropriate AI-generated content can negatively impact users. This project focused on ensuring that AI-generated content was safe, unbiased, and aligned with ethical guidelines.

My contributions included:

- a) Identifying and flagging biased or inappropriate content generated by AI models.
- b) Reviewing AI-generated advertisements, search results, and news articles for misleading or offensive material.
- c) Ensuring AI models adhered to ethical content guidelines, preventing misinformation and harmful outputs.

- d) Analyzing how AI models classified and filtered sensitive topics, ensuring fair and unbiased content moderation.

Through this project, I developed a deeper understanding of AI ethics, fairness in machine learning, and the impact of content moderation on user trust. Additionally, I gained insight into how AI bias can emerge from training data and algorithmic decision-making, emphasizing the need for continuous evaluation and refinement of AI models to ensure inclusivity and fairness.

Skills Gained and Contribution

Throughout my internship at Peroptyx, I acquired a diverse set of skills spanning technical expertise, professional communication, and critical problem-solving. The experience allowed me to apply my academic knowledge in AI and machine learning to real-world evaluation projects, enhancing my overall competency in AI model assessment, data validation, and content moderation.

1. Technical Skills

During the internship, I gained hands-on experience in:

- a) **AI Model Evaluation:** Assessing AI-generated search results, geospatial data accuracy, and content moderation outputs to ensure quality and relevance.
- b) **Data Annotation and Labeling:** Understanding the significance of high-quality labeled data for supervised machine learning models and contributing to structured annotation processes.
- c) **Geospatial Data Analysis:** Verifying mapping datasets, identifying inconsistencies, and ensuring accurate location-based AI recommendations.
- d) **Search Engine Optimization (SEO) & NLP:** Evaluating search engine outputs and understanding how AI-based personalization and natural language processing affect ranking and user intent.
- e) **Bias Detection & Ethical AI Practices:** Identifying potential biases in AI-generated content and ensuring adherence to ethical AI principles in data evaluation.
- f) **Data Quality Assurance:** Implementing human-in-the-loop methodologies to validate AI predictions, reducing errors, and improving overall model performance.

These technical skills have significantly strengthened my understanding of how AI models operate, their limitations, and how human feedback can refine their effectiveness. In the future of a young machine learning engineer these skills and knowledge would come to be very efficient and unexpandable.

2. Professional and Soft Skills

Beyond technical expertise, the internship helped me develop essential professional and interpersonal skills, including:

- a) **Analytical Thinking:** Evaluating AI-generated outputs critically, identifying inconsistencies, and providing constructive feedback.
- b) **Attention to Detail:** Ensuring accuracy in search results, geospatial data, and content moderation tasks to maintain high-quality AI evaluation.
- c) **Time Management:** Balancing multiple projects efficiently while adhering to deadlines and task completion guidelines.
- d) **Collaboration & Communication:** Coordinating with team members, sharing insights, and understanding project requirements effectively.
- e) **Adaptability:** Learning and adjusting to evolving AI evaluation methodologies, working on diverse projects, and applying new knowledge dynamically.
- f) **Problem-Solving:** Identifying AI-generated inconsistencies and proposing solutions to improve data quality, content filtering, and search accuracy.

These soft skills have significantly enhanced my ability to work in a dynamic AI-driven environment, improving my efficiency and professional communication abilities.

Conclusion and Recommendations

The Peroptyx internship provided a valuable learning experience, exposing me to real-world AI applications and evaluation challenges. Through AI search result assessment, geospatial data validation, and bias detection, I gained deep insights into data quality, AI ethics, and machine learning performance metrics.

1. Summary of Experience

- a) Worked on AI model evaluation, ensuring accuracy, fairness, and relevance of AI-generated content.
- b) Developed expertise in geospatial data verification, search engine optimization, and content moderation.
- c) Acquired practical experience in human-in-the-loop AI validation, an essential component in AI model training.
- d) Strengthened both technical and professional skills, improving analytical thinking, collaboration, and problem-solving.

This internship has been a crucial step in bridging my academic knowledge with industry applications, allowing me to understand how AI models interact with real-world data and user experiences.

2. Future Scope and Career Impact

The skills and knowledge gained during this internship will significantly impact my career in AI, data science, and machine learning. The ability to evaluate AI models, detect biases, and improve data annotation quality is essential for future roles in:

- a) AI Research & Development: Enhancing AI models through iterative evaluation and ethical AI training.
- b) Machine Learning Engineering: Applying data preprocessing, model tuning, and quality assurance techniques for AI improvement.

- c) Search Engine Optimization (SEO) & NLP: Contributing to search algorithm improvements and natural language processing applications.
- d) Geospatial Data Science: Working on mapping technologies and AI-driven navigation enhancements.

The experience has reinforced my interest in AI ethics, fairness, and model interpretability, which will be pivotal in my future AI and ML career pursuits.

3. Recommendations for Future Interns

For future interns joining Peroptyx, I recommend the following:

- a) Build a strong foundation in AI & data science: Understanding machine learning fundamentals will help in AI model evaluation tasks.
- b) Develop attention to detail: AI model evaluation requires precision in identifying inconsistencies, biases, and errors.
- c) Be proactive in learning: AI evaluation methodologies evolve, so staying updated with industry trends is crucial.
- d) Manage time effectively: Working on multiple AI projects requires strong time management skills to meet evaluation deadlines.
- e) Communicate effectively: Sharing feedback, discussing findings, and understanding project goals are key to delivering high-quality evaluation results.

This internship is an excellent opportunity to gain hands-on experience in AI model evaluation and data annotation, making it a valuable stepping stone for a career in artificial intelligence and machine learning.